

Git Practice Exercises (Commands Guide)

Exercise 1: Basic Git Workflow

- `mkdir git-practice && cd git-practice`
- `git init`
- `echo "# Git Practice" > README.md`
- `git add README.md`
- `git commit -m "Initial commit with README"`
- `echo "Learning Git basics" >> README.md`
- `git add README.md`
- `git commit -m "Update README"`
- `git log`

Exercise 2: Branching and Merging

- `git checkout -b feature`
- `touch feature.txt`
- `git add feature.txt`
- `git commit -m "Add feature file"`
- `git checkout main`
- `git merge feature`
- `git branch -d feature`

Exercise 3: Conflict Resolution

- `git checkout -b branch1`
- `echo "Line from branch1" > conflict.txt`
- `git add conflict.txt && git commit -m "Branch1 change"`
- `git checkout main`
- `git checkout -b branch2`
- `echo "Line from branch2" > conflict.txt`
- `git add conflict.txt && git commit -m "Branch2 change"`
- `git checkout main`
- `git merge branch1`
- `git merge branch2`
- Resolve conflict manually and commit

Exercise 4: GitHub Collaboration

- Fork repository on GitHub
- `git clone`
- `git checkout -b my-changes`
- `git add .`
- `git commit -m "My changes"`
- `git push origin my-changes`
- Create Pull Request on GitHub

Exercise 5: Advanced Workflow

- `git init`
- `touch index.html app.js style.css`
- `git add . && git commit -m "Initial project files"`
- `echo "node_modules/" > .gitignore`
- `git add .gitignore && git commit -m "Add gitignore"`
- `git stash`
- `git stash pop`
- `git tag -a v1.0.0 -m "First release"`
- `git push origin main`
- `git push origin --tags`